

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

In Cutting Edge, learn more about the mysterious particles known as Axions. Then in Ignorance, find out if you can outrun a T-Rex. Finally, delve into the history of the Solar System in Space Age.



Polymer wafer for vaccine delivery

A novel method of delivering vaccines could allow for much widespread distribution, and consists of a polymer wafer that is stuck beneath the tongue.

<http://dgit.in/polywaf>

WHAT'S NEW

Drones team up to lug it

The drones that are being developed for delivery of parcels are being designed to carry payloads of around 2.5

kilograms. However, that does not mean that heavier parcels have to be relegated to the slow delivery mechanisms by parcel bikes or vans. If a single quadcopter cannot lift a load, maybe a bunch of them can work together to deliver a heavier parcel. To check this idea, researchers from Georgia Tech developed a modular solution that allows small fleets of autonomous drones to work together to deliver larger packages. The system is dynamic and can keep scaling for heavier loads, as long as there is some room available for the drone to grab on. Entire swarms

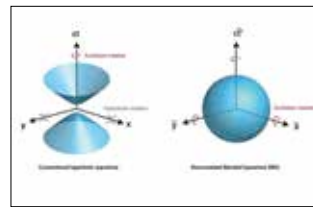


currently being developed by parcel companies such as UPS, where a delivery van acts as a base station for drones. The way the system works is that the delivery van drives around, making the deliveries, while drones deliver additional parcels and charge up between deliveries. In such a setup, the drone swarm would be able to smartly adapt to the requirements of the particular delivery.

<http://dgit.in/drnpcck>

Spacetime crystals

When Einstein came up with his theories of relativity, he proposed a new way of looking at the universe around us, where three space dimensions and a fourth time dimension are melded together in a four dimensional spacetime. The problem with this four dimensional spacetime is that a lot of the equations require adding a negative sign to the time value, a correction that is not required for the



space values. The consequence of this is that physicists cannot work with four dimensional spacetime using traditional Euclidean geometry, and have to instead have to use hyperbolic geometry. A Penn

State scientist, Venkatraman Gopalan, has provided a new two step approach that allows physicists to deal with spacetime where the difference between the space dimensions and the time dimension are better reconciled. The research has the potential to bridge the gap between quantum mechanics and gravity, providing a more coherent explanation of how the world works.

<http://dgit.in/stcrystals>



Tom Brock dies at 94

The scientist who kickstarted a whole new field of extremophile biology passed away after a fall. Brock discovered the bacteria *Thermus aquaticus*, and the enzyme inside that allowed it to thrive everywhere, that in turn is a key ingredient in the PCR tests used to combat the outbreak.

<http://dgit.in/tombrock>



Combating the Marburg virus

There have been no vaccines or antiviral treatments for the deadly Marburg virus, which is similar to Ebola and transmitted by the African Fruit Bat to humans. Researchers have now identified an antiviral that prevents the virus from leaving the cells it has infected.

<http://dgit.in/marburgv>



Another coronavirus made the jump

A team of researchers have identified a new variant of the coronavirus, named CCoV-HuPn-2018 that made the jump from dogs to humans in Malaysia in 2018. Further studies can confirm that this is the eighth known coronavirus to cause illnesses in people.

<http://dgit.in/hupn18>